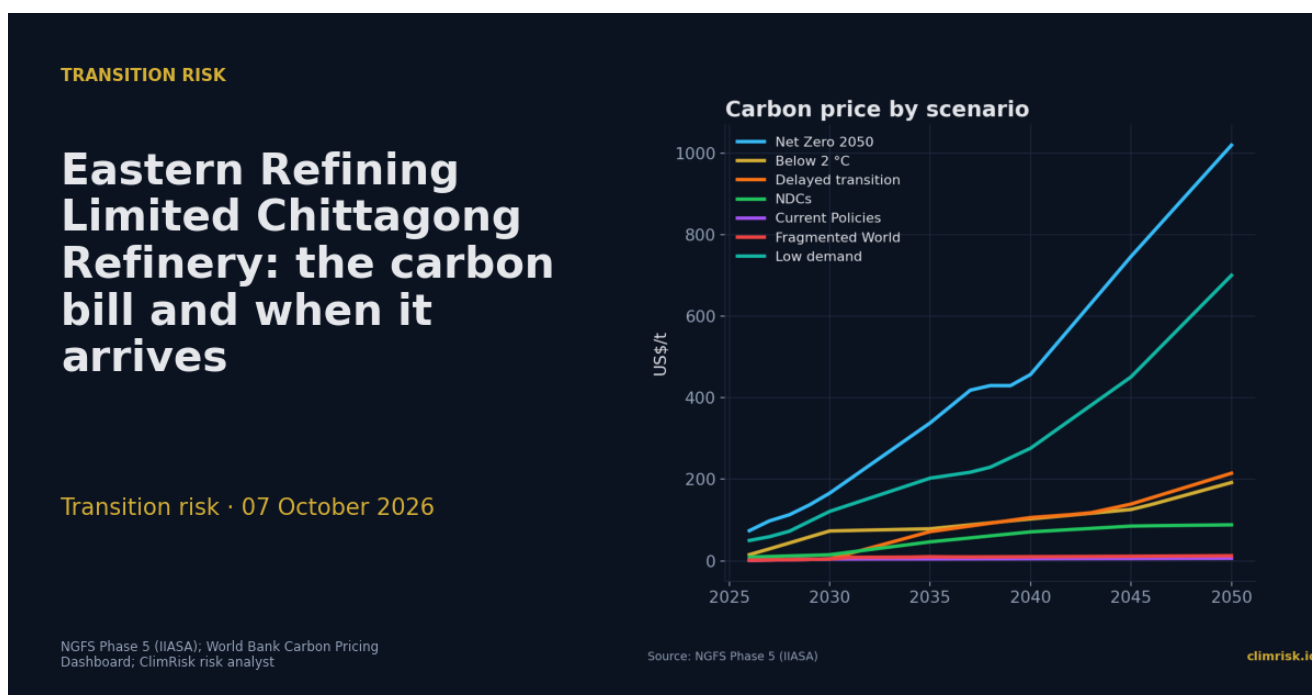


Eastern Refining Limited Chittagong Refinery: the carbon bill and when it arrives

Eastern Refining Limited Chittagong Refinery emits 163 kt CO₂e a year. The carbon price it pays today is \$0/t; under Net Zero 2050 the modelled price passes \$1,019/t by 2050. Present-value cost to 2050 runs from \$4.1m (current policies) to \$90.1m (Net Zero 2050) [1].

ClimRisk risk analyst · 2026-10-07



JAS 2026. The Oceanic Niño Index reads +2.16 °C: El Niño. Since 1950 it has crossed +2 °C only 4 times [2]. The ocean is the loud part of the climate story this year. What follows is about the part that does not make the news: what climate policy and climate hazards do to the value of a specific set of industrial assets.

Eastern Refining Limited Chittagong Refinery in six numbers

The carbon bill, as modelled

163 kt

CO₂e a year

Climate TRACE v6 register

\$500.0m

asset value

user-supplied value

\$0/t

carbon price today

World Bank Carbon Pricing Dashboard

\$4.1m

transition cost, current policies

Monte Carlo mean, PV 2026–2050

\$90.1m

transition cost, Net Zero 2050

carbon + abatement + stranding

\$38.8m

stranded value, Net Zero 2050

output falls below the run-rate

Source: ClimRisk risk analyst; NGFS Phase 5 (IIASA); World Bank Carbon Pricing Dashboard

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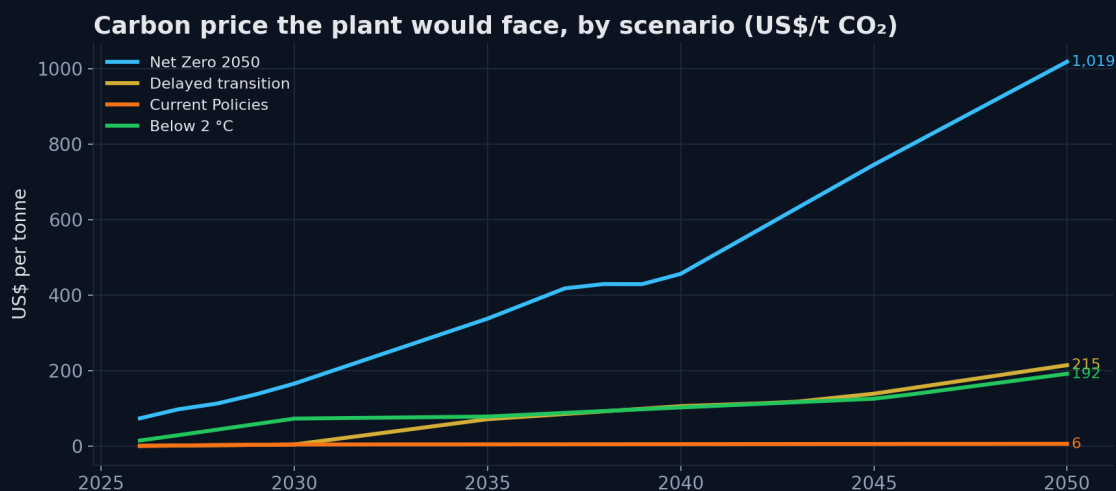
Each number is a Monte Carlo mean or an input from the register; sources on the image.

21.8× between scenarios

The same plant costs \$4.1m under current policies and \$90.1m under Net Zero 2050. Under the Net Zero 2050 scenario the transition cost (\$90.1m) is about the same as the modelled physical damage (\$90.6m).

The loud risk: the carbon price path

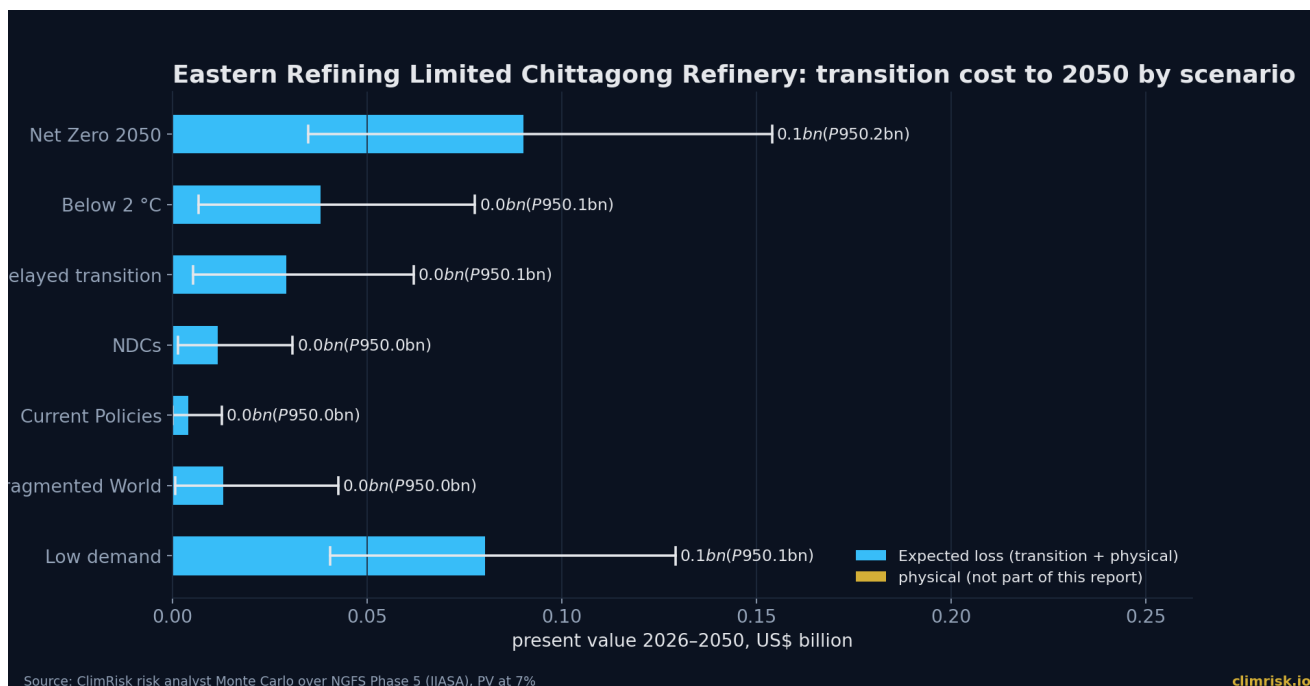
Carbon pricing is the visible part. Today 0 instrument(s) in force reach this asset, with an effective price of \$0/t [3]. The scenario paths are what changes: Net Zero 2050 \$74/t in 2026 to \$1,019/t in 2050; Delayed transition \$1/t in 2026 to \$215/t in 2050; Current Policies \$1/t in 2026 to \$6/t in 2050 [1].



Source: NGFS Phase 5 (IIASA), median of GCAM / MESSAGEix / REMIND, regionalised; ClimRisk engine

climrisk.io

Median carbon price by scenario for this asset's region [1].



Carbon cost paid or abated plus stranded value, with the 5th to 95th percentile [1].

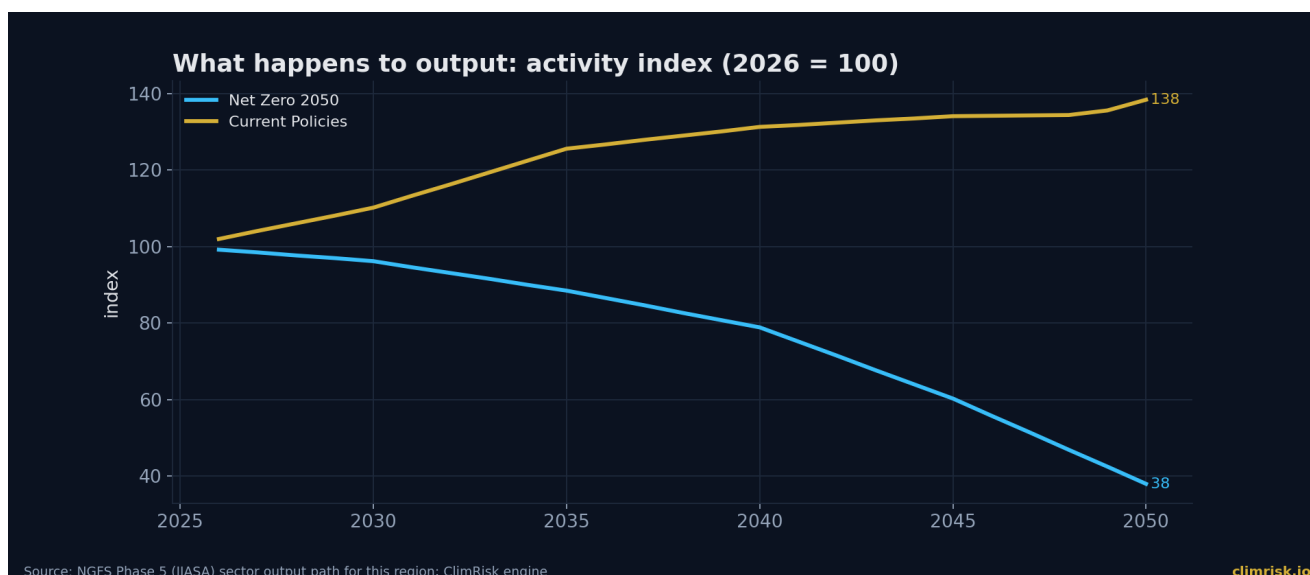
Scenario	Carbon cost PV (no action)	Cost with abatement response	Stranded value	Transition total
Net Zero 2050	\$69.6m	\$51.3m	\$38.8m	\$90.1m
Below 2°C	\$28.3m	\$23.4m	\$14.6m	\$38.0m
Delayed transition	\$20.4m	\$17.1m	\$12.1m	\$29.3m
Nationally Determined Contributions (NDCs)	\$12.2m	\$11.7m	\$0	\$11.7m
Current Policies	\$1.2m	\$1.2m	\$2.9m	\$4.1m
Fragmented World	\$3.8m	\$3.8m	\$9.3m	\$13.0m
Low demand	\$44.7m	\$33.9m	\$46.3m	\$80.3m

Present value 2026–2050, mean of the Monte Carlo draws [1].

The quiet risk: which policy actually reaches the asset

No carbon price currently applies to this asset's sector in its jurisdiction. [3]

Where the emissions sit decides which policies reach the site.



Output under each scenario, from the NGFS model path for the asset's sector and region. Stranded value is what the falling run-rate leaves behind [1].

What the operator can do about it

Lever	Share of emissions	Marginal cost (US\$/t)	Available from	Basis
Efficiency, electrification, fuel-gas switching	5%–10%–15%	0–20–60	2026	analyst judgement
Capture on hydrogen units and catalytic cracking	25%–40%–55%	40–80–120	2030	published range

Library ranges unless the operator supplies its own levers. Get site quotes and rerun.

Under Net Zero 2050 the model abates when the price reaches each lever's cost. Without any abatement the mean total loss is \$196.7m, against \$178.9m with the sector levers: \$17.9m avoided.

Carbon cost to 2050 (delayed transition, present value) \$20.4m if nothing is done (4% of asset value); with the sector's abatement levers deployed as prices rise \$17.1m (\$3.2m avoided). Electrify process heat, add CCS on hydrogen/SMR units, and plan throughput reduction in line with falling oil demand.

Modelled response: Efficiency, electrification, fuel-gas switching starts about 2035 (95% of runs by 2040); Capture on hydrogen units and catalytic cracking starts about 2044 (7% of runs by 2040). Get site-specific quotes for these levers (cost per tonne, lead time) and rerun with them in place of the library ranges; start permitting and procurement now for the long-lead options.

What this means, by reader

For the operator. The cost depends on timing as much as price: abatement started late is paid for at the highest prices.

For the lender or investor. The question is which scenario the plan sits in, and whether the stranded-value row (\$38.8m under Net Zero) is covered by the asset's remaining life.

Method and limitations

A Monte Carlo over NGFS Phase 5 carbon-price and sector-output paths [1], regionalised to the asset and run for seven scenarios. The carbon cost is emissions times the price the plant faces, passed through to customers in part; abatement levers switch on when the price reaches their marginal cost; stranded value is the asset value lost when output falls below the run-rate that justified it. abated when the carbon price the plant faces reaches a lever's levelised cost (sector levers, lead times, irreversible); the unabated cost is shown as cost of inaction
Screening-grade model: validate asset values, defence standards and emissions with the owner before decisions.
Direct physical damage only — business interruption and supply-chain losses are not included.

How ClimRisk measured this

- Register → asset, emissions, capacity and value basis.
- Policy → World Bank instruments in force and pipeline for the jurisdiction and sector.
- Scenarios → NGFS Phase 5 price and output paths, three models, seven scenarios.
- Monte Carlo → draws on price, pass-through, charged share and value; discounted at 7%.
- Abatement → sector lever library with marginal costs; the plant abates when the price crosses each one.

Sources

1. NGFS Phase 5 climate scenarios (IIASA Scenario Explorer) — carbon price and sector output, 7 scenarios, 3 models
<https://data.ene.iiasa.ac.at/ngfs/>
2. NOAA Climate Prediction Center, Oceanic Niño Index (ERSSTv5)
https://www.cpc.ncep.noaa.gov/products/analysis_monitoring/ensostuff/ONI_v5.php
3. World Bank Carbon Pricing Dashboard — instruments in force and pipeline <https://carbonpricingdashboard.worldbank.org/>

Produced by the ClimRisk engine from the sources listed. Model outputs with the limitations stated; not forecasts or investment advice.